OmicsLite BRSP | Assignment Week 2

Differential Expression Analysis (GEO2R) Practice with GSE3053 Dataset

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Background

One of the pressing issues in the world is related to food production in the climate crisis. The sea water level that keeps increasing because of global warming can contaminate the agricultural land located close to the shore, especially in an archipelago state like Indonesia. Meanwhile, rice (*Oryza sativa*), as the most popular and widely consumed staple food in Asia, requires lots of fresh water to grow. While controlling salt intrusion in fresh water is really complex, researchers have been looking for the underlying mechanism of several rice genotypes that developed salt-tolerance traits.

NCBI gene expression omnibus (GEO) is one of public data repositories whose database can be mined to conduct differentially gene expression (DEG) analysis from various transcriptomics research. Embedded within NCBI GEO is GEO2R which is a web-based tool to analyze differential expression from available data and generate plots to visualize the relation of two or more sample groupings. In the previous study by Walia et al. (2005), two *indica* genotypes', FL478 and IR29, crown and growing point tissue were investigated for expression analysis in control and salt stress treatment. Rice has a range of salt-tolerance variation within the cultivated gene pool despite being a salt-sensitive species. FL478 is a recombinant inbred line developed as salt-tolerant genotype while IR29 is a sensitive parent population. Affymetrix Rice Genome arrays (GPL2025) was utilized as the platform to determine expression profiles.

In this study, the dataset with the GEO ID of GSE3053 will be explored to practice navigating NCBI GEO and interpreting the results of DEG analysis with GEO2R.

Methods

The dataset was determined according to the pipeline in Figure 1.

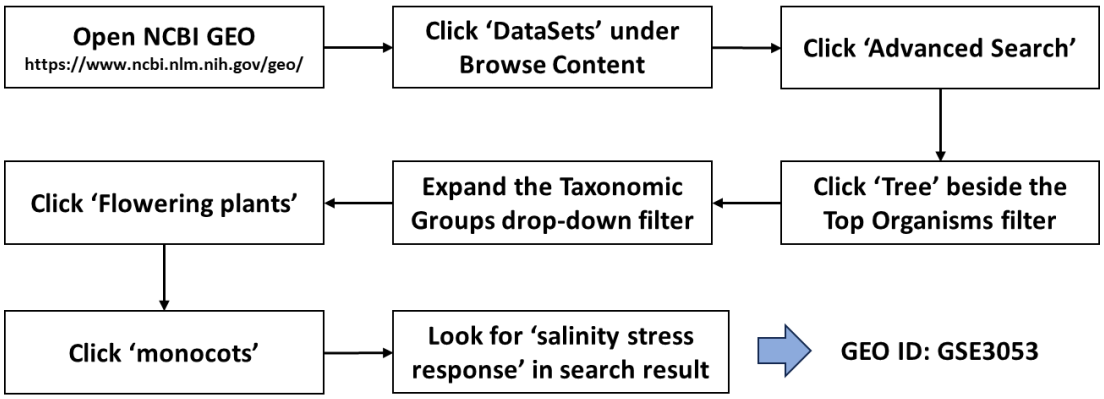

Figure 1. The pipeline to explore and determine NCBI GEO dataset.

The GSE3053 dataset was analyzed for differential expression using GEO2R. The information regarding all samples in the dataset can be found through the summary and research paper. There are four sample groupings: Control FL (3 replicates of FL478 control samples), Treated FL (3 replicates of FL478 salt-stressed samples), Control IR (2 replicates of IR29 control samples), and Treated IR (3 replicates of IR29 salt-stressed samples). Under the options selection for parameters, Benjamini & Hochberg was applied as default to adjust the p-value of 0.05 as significance level cut-off for false discovery rate in multiple testing. Force normalization was turned on to eliminate technical bias and improve accuracy. Log2 fold change (FC) threshold was kept 0 as default. The volcano and mean-difference plot contrasts were set for Control FL vs Treated FL and Control IR vs Treated IR alternately. The volcano plot results from each grouping's comparison were explored and the significant genes were downloaded.

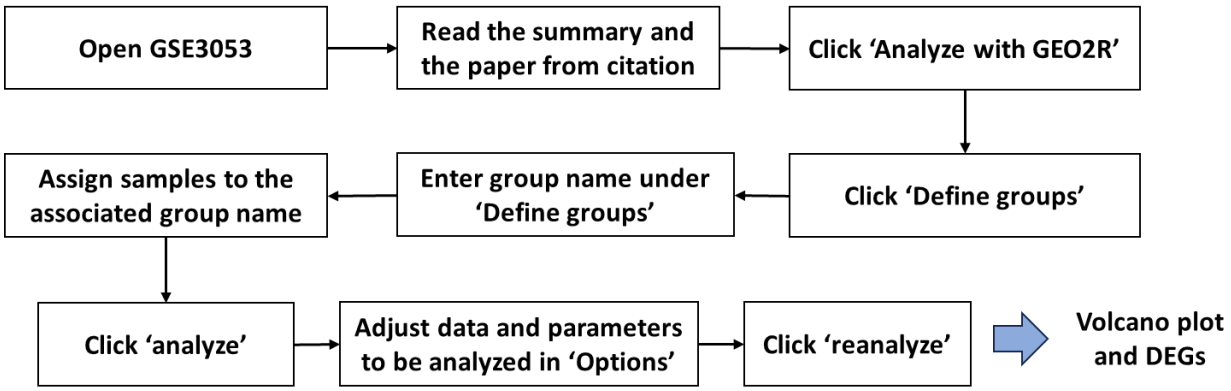

Figure 2. The pipeline to analyze GSE3053 dataset and generate results.

The upregulated and downregulated genes were determined and analyzed for GO terms and KEGG pathway using g:Profiler (https://biit.cs.ut.ee/gprofiler/gost) to elucidate its relation with biological processes. The methods were replicated by following the designated study pipelines and its parameters three times to check the research reproducibility.

Result

From NCBI GEO dataset exploration, the GSE3053 dataset was found to be in line with the author's research topic interest. Therefore, the author proceeded with GSE3053 to practice explaining DEGs analysis results from GEO2R. Two DEGs were detected from the volcano plot of Control FL vs Treated FL, with one gene being upregulated and the other downregulated. Moreover, ten DEGs were detected from the volcano plot of Control IR vs Treated IR, with two genes being upregulated and seven downregulated (Figure 3). The only GO term that was found was one gene from FL478, with the ID of OsAffx.28184.2.S1_at which is associated with UDP-xylosyltransferase activity (GO:0035252) and xylosyltransferase activity (GO:0042285). Xylotransferase activity is involved in plant growth and development under abiotic stress (Takano et al., 2015).

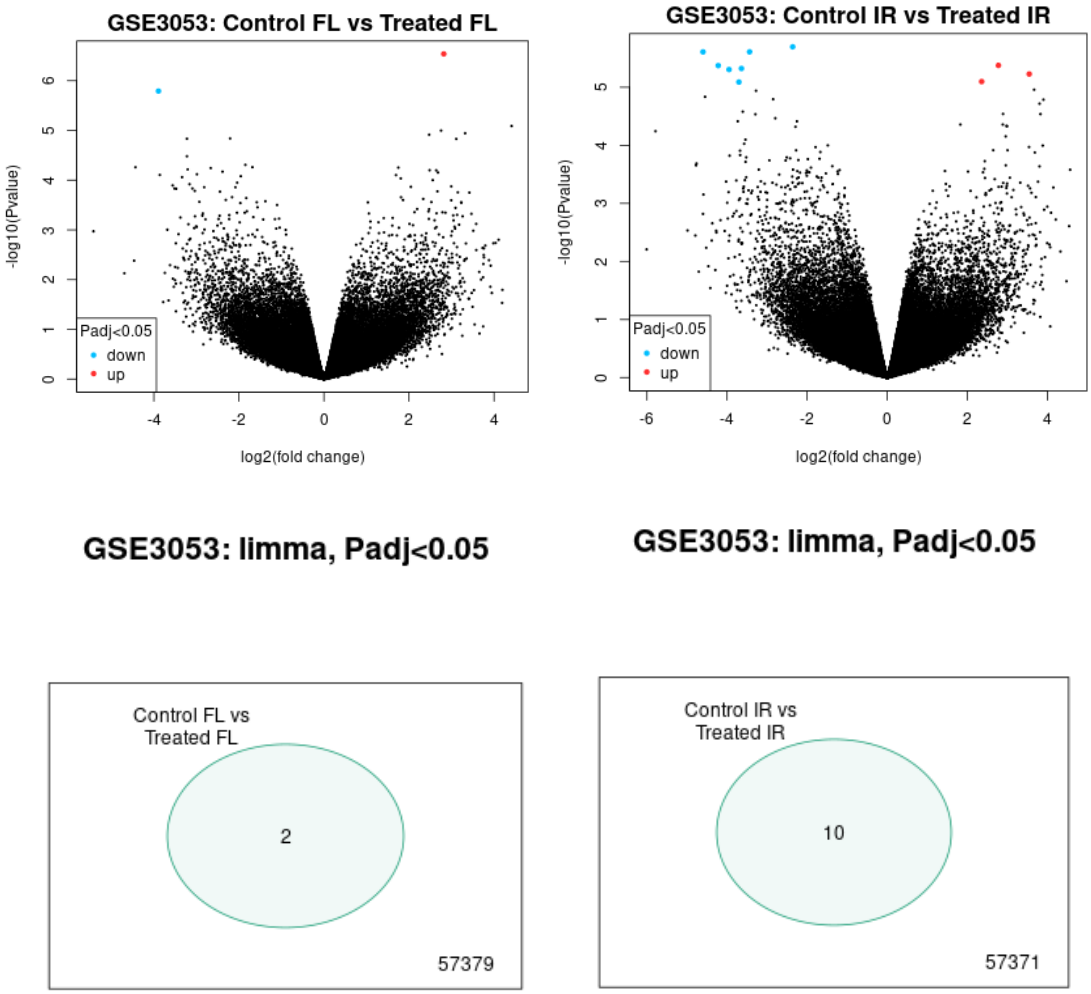

Figure 3. Volcano plot and diagram venn result of GSE3053 datasets analyzed with GEO2R.

Conclusion

In response to salt stress treatment the salt-sensitive rice genotype, IR29, mechanism of action is to upregulate two genes and downregulate seven genes whose biological process association is still unknown. Meanwhile, in salt-tolerant rice genotype, FL478, the mechanism of action is to upregulate one gene related to xylotransferase activity and downregulate one gene with unknown biological process association. The replication showed consistent results in three runs. In conclusion, the author has been familiarized in navigating NCBI GEO and interpreting DEGs results from GEO2R with GSE3053 datasets.

References

Walia H, Wilson C, Condamine P, Liu X et al. Comparative transcriptional profiling of two contrasting rice genotypes under salinity stress during the vegetative growth stage. *Plant Physiol* 2005 Oct;139(2):822-35. PMID: 16183841

Takano, S., Matsuda, S., Funabiki, A., Furukawa, J., Yamauchi, T., Tokuji, Y., Nakazono, M., Shinohara, Y., Takamure, I., & Kato, K. (2015). The rice RCN11 gene encodes β1,2-xylosyltransferase and is required for plant responses to abiotic stresses and phytohormones. Plant Science, 236, 75–88. https://doi.org/10.1016/j.plantsci.2015.03.022